

Automatic design of missing and damaged teeth. A brief for dental laboratories.

What it does

ToothCraft is a neural network that takes a scan of a dental arch and designs the tooth that is missing or damaged, shaped to that specific patient, fitted to the neighbouring teeth, and returned as an STL in the coordinate system of your scan.

You send the scan and the tooth number; you get back a mesh you can open in your CAD software. There is nothing to install and no change to how you scan.

Full crowns

Inlays

Onlays / overlays

Missing teeth

Upper & lower jaw



A real case: a molar prepared for a crown. ToothCraft sees only the left column. The middle column is what it generated; the right column is what a technician designed for the same case.

Where it stands today

What it gets right

- Shape, size and position of the tooth
- Contact with both neighbouring teeth
- Overall anatomy of the occlusal surface
- Occlusion, when the opposing jaw is supplied
- A watertight mesh, aligned to your scan

What it does not do yet

- **Fine surface detail.** Fissures and ridges come out smoothed, so the output is a starting shape for the technician rather than a finished restoration.
- **Run end to end on its own.** The network produces the tooth in seconds. The manual preparation and export around it add a few minutes per case. Automating that is routine work, set up once per scanner and export format.

How it works, briefly

ToothCraft is a **neural network**, so it learns tooth design from examples rather than from rules. It was trained on thousands of complete arches from which teeth were removed automatically, so that for every example the correct answer was known. Given a new scan, it looks only at the local region around the target tooth, roughly twice the size of the tooth and including both neighbours, and generates the

missing surface. The generative model is of the same family as current image generators. The surface is then converted back into a triangle mesh.

Published as: Pukanec D., Kubík T., Španěl M. *From Synthetic Data to Real Restorations: Diffusion Model for Patient-specific Dental Crown Completion*. VISAPP 2026, pp. 734–742. Code and trained models are public.

What we are looking for

We are a university research group, not a vendor. The honest picture of where we are stuck, and what a laboratory can do about it:

What we lack

The public dental datasets we train on contain teeth whose surfaces are already smoothed, with the fine anatomy largely washed out. A model cannot learn detail that is not in its training data, and this single factor is the main reason our output is still short of a finished restoration. What does not exist publicly is **paired data from practice**: the scan as it arrived, together with the design a technician actually produced for it.

What we offer

- **A trial batch of your cases processed at no cost**, with results returned as STL, so you can judge the method on your own work.
- **Early access** to the method as it improves, before anything is productised.
- **Tuning towards your workflow**, meaning the export formats and conventions you already use.
- **A say in what we work on next**, since the problems you hit in practice are the ones worth solving.

What we need

- **Scans with the finished design** for the same case, the pairs described above. Even a few dozen would change what we can train.
- **Your judgement on the output**. Where would a technician accept this shape, and where would it be faster to start from scratch?
- **A sense of where this would fit** in your process, and what it would have to do to be worth using.

What to send us

Nothing beyond a standard intraoral scanner export is needed.

Required, for each case

- **The arch scan as `.stl` or `.obj`**
The whole arch, or at least the region around the tooth including both neighbouring teeth.
- **Which tooth** is to be designed
The tooth number (e.g. 36) is enough.
- **Which jaw**: upper or lower

Optional, but it helps

- **The opposing jaw scan**
Lets the method shape the occlusal surface to the bite.
- **Tooth and gingiva labels**
Which part of the mesh is which tooth. Saves us a manual step.
- **The finished design for the same case**
This is what we need most; see “What we are looking for” above.

If you can export labels

One label per mesh vertex, in any consistent scheme, since we will adapt to what your software produces. A minimal example of what travels with a case:

```
{
  "jaw": "lower",
  "target_tooth": 36,
  "labels": [0, 0, 36, 36, 37, 0, ...]  // 0 = gingiva, tooth number = tooth
}
```

Practicalities

- **How to send it:** email, or a download link for large files (any cloud share).
- **Turnaround:** the network runs in seconds and the manual steps add a few minutes per case, so a batch comes back within days.
- **Cost:** a trial batch is free, because it is a demonstration and your feedback is worth more to us than a fee. Use at production volume is a different matter: it runs through the **Czech AI Factory (CZAI)** programme, which you would join as a partner. Ask us and we will set out what that involves before you commit to anything.

How we handle your data

- Scans you send are used **only to process your cases**.
- We will **not train our models** on them, and **not publish them**, unless you explicitly allow it. If you do, we will agree in writing what that covers.
- Nothing we return or publish carries patient identifiers; the data we work with is geometry and a tooth number.
- If you would rather have a formal agreement (NDA or a data-processing agreement) before sending anything, tell us and we will arrange it through the university.

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Examples and 3D viewer: <https://ikarus1211.github.io/toothcraft/>

Code: github.com/ikarus1211/VISAPP_ToothCraft

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